Lecture-1
04.10.2026

3.6 mark

1, 2, 3, 4, 5

DFT → Discrete Fourier Transform

FFT → Fast Fourier Transform

Discrete Fourier Transform

Fourier Transform converts :

time domain signal ⟶ frequency Domain signal

x, y X, Y

DFT eqn: $$X(f) = \int_{-\infty}^{+\infty} x(t)\, e^{-j2\pi ft}\, dt \qquad \boxed{f = \frac{K}{N}}$$

$$X(K) = \sum_{-\infty}^{+\infty} x(n)\, e^{\frac{-j2\pi Kn}{N}}$$

$$\boxed{X(K) = x_N W_N^{Kn}}$$

Twiddle factor

Here, Kn → column, row

$$W_N = e^{-j2\pi \frac{Kn}{N}}$$

K ↑ , → n

[ ] Kxn

Lecture - 2

11.10.2026

Project proposal DSP 3208

ECG → Mode detection to arythimia detection.

simulink project / publish করতে হবে)

use real ECG data.

Lecture - 3

18.10.2026

Q Perform DFT of the following sequences:

$$x(n) = \{1, j, -1, -j\}$$

Sol^n: We know,

$$X(k) = \sum_{N=0}^{N-1} x(n) e^{-j 2\pi k n / N}$$

$$= \sum_{0}^{4-1} x(n) \cdot e^{-\frac{j 2\pi k n}{4}}$$

$$= \sum_{0}^{3} x(n) \cdot e^{-j 2\pi k n / 4}$$

When,

$K = 0$

$$X(0) = \sum_{0}^{3} x(n)\, e^{-j2\pi n \cdot 0/4}$$

$$= x(0) + x(1) + x(2) + x(3)$$

$$= 1 + j - 1 - j$$

$$= 0$$

$e^{-j\pi}$

$= \cos\pi - j\sin\pi$

$= -1 - j.0$

$= -1$

When $K = 1$,

$$X(1) = \sum_{0}^{3} x(n) \cdot e^{-j2\pi n 1/4}$$

$$= \sum_{0}^{3} (x) \cdot e^{-j\frac{\pi}{2} n}$$

$e^{-j.\pi/2}$

$= \cos\frac{\pi}{2} - j\sin\frac{\pi}{2}$

$= 0 - j.1$

$= -j$

$$= x(0) \cdot e^{-j\frac{\pi}{2} \cdot 0} + x(1)\, e^{-j.\pi/2 \cdot 1} + x(2) \cdot e^{-j\pi/2 \cdot 2} + x(3)\, e^{-j\pi/2 \cdot 3}$$

$$= 1.1 + j(-j) + (-1)(-1) + (-j)(j)$$

$$= 1 - j^2 + 1 - j^2$$

$$= 1 + 1 + 1 + 1$$

$$= 4$$

$j^2 = -1$

$j = \sqrt{-1}$

$e^{-j\frac{3\pi}{2}}$

$= \cos\frac{3\pi}{2} - j\sin\frac{3\pi}{2}$

$= 0 - j(-1)$

$= j$

When $K=2$,

$$X(2) = \sum_{n=0}^{3} x(n) \cdot e^{-j2\pi 2n/4}$$

$$= \sum_{n=0}^{3} x(n) \cdot e^{-j\pi n}$$

$$= x(0) \cdot e^{-j\pi 0} + x(1) e^{-j\pi 1} + x(2) \cdot e^{-j\pi 2} + x(3) e^{-j\pi 3}$$

$$= 1 + j(-1) + (-1)(-j) + (-j)(j)$$

$$= 1 - j + j - (j)^2$$

$$= 0$$

$K=3$,

$$X(3) = \sum_{n=0}^{3} x(n) e^{-j2\pi 3n/4}$$

$$= \sum_{n=0}^{3} x(n) e^{-j\pi n 3/2}$$

$$= x(0) e^{-j\pi 0 3/2} + x(1) e^{-j\pi 1 3/2} + x(2) e^{-j\pi 2 3/2} + x(3) e^{-j\pi 3 3/2}$$

$$= 1 + j^2 + 1 + j^2 = 1 - 1 + 1 - 1 = 0$$

$\therefore X(k) = \{0, 4, 0, 0\}$

Exam - 4 Twiddle method - 4

We know,

$$X(k) = \sum_{n=0}^{N-1} x(n)\, e^{-j2\pi kn/N}$$

$$= W_N^{kn} \cdot x_N$$

$$= W_4^{kn} \cdot x_4$$

$$= \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{bmatrix} W_4^0 & W_4^1 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W_4^6 \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix} \end{array} \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ j \\ -1 \\ -j \end{bmatrix} \Rightarrow \begin{bmatrix} 1+j-1-j \\ 1-j^2+1-j^2 \\ 1-j-1+j \\ 1+j^2+1+j^2 \end{bmatrix}$$

$\Rightarrow -j^2 j^4 = -1$

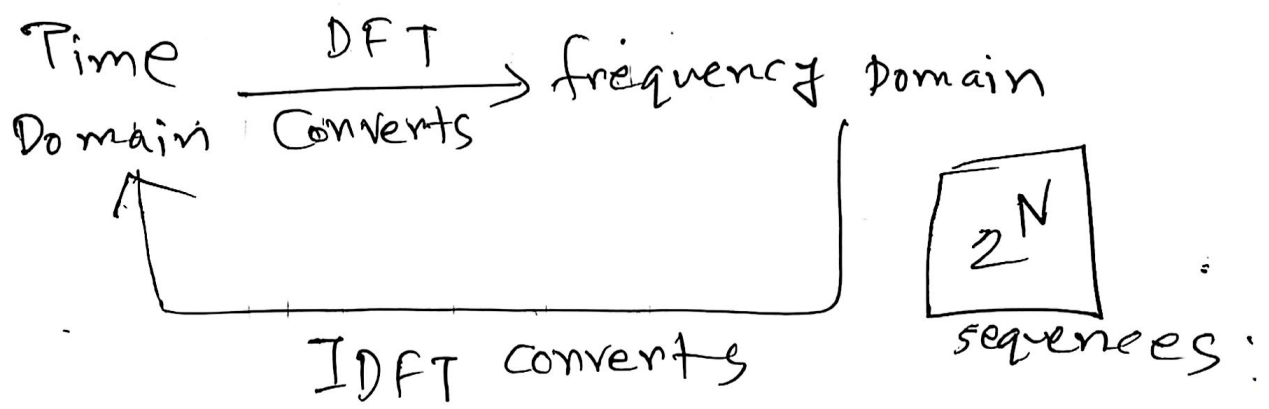
$= -j^2 = 1$

$$= \begin{bmatrix} 1+j-1-j \\ 1+1+1+1 \\ 1-j-1+j \\ 1+j^4+1+j^4 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ 0 \\ 0 \end{bmatrix}$$ ✓

Lecture-4

28.10.2026

IDFT (Inverse Discrete fourier Transform)



$x(n) \xrightarrow{DFT} X(k)$

$X(k) \xrightarrow{IDFT} x(n)$

Formula:

$$x(n) = \frac{1}{N}\sum_{K=0}^{N-1} X(K)\, e^{j 2\pi kn/N}$$

Q. Perform IDFT of the following sequence:

$$X(K) = \{0, 4, 0, 0\}$$

when $n=0$;

$$x(0) = \frac{1}{4}\sum_{K=0}^{3} X(K)\, e^{j 2\pi \cdot 0 \cdot K/4}$$

$$= \frac{1}{4}\sum_{K=0}^{3} X(K)$$

$$= \frac{1}{4}\{X(0) + X(1) + X(2) + X(3)\}$$

$$= \frac{1}{4}\{0 + 4 + 0 + 0\}$$

$$= 1$$

when $n=1$,

$$x(1) = \frac{1}{4}\sum_{K=0}^{3} X(K) e^{j 2\pi K/4}$$

$$= \frac{1}{4}\{X(0)e^{0} + X(1)\, e^{j2\pi/4} + X(2)e^{j4\pi/4}$$

$$+ X(3)\, e^{j 2\pi 3/4}$$

$$= \frac{1}{4}(0 + 4e^{j2\pi/4} + 0 + 0)$$

$$= e^{j2\pi/4}$$

$$= j$$

when $n = 2$,

$$x(2) = \frac{1}{4}\sum_{K=0}^{3} X(K)\, e^{j2\pi K 2/4}$$

$$= \frac{1}{4}\{X(0)\, e^{j2\pi 02/4} + X(1) e^{j2\pi 2/4} + X(2) e^{j2\pi 22/4} + X(3) e^{j2\pi 32/4}\}$$

$$= \frac{1}{4}\{0 + 4e^{j4\pi/4} + 0 + 0\}$$

$$= e^{j\pi}$$

$$= \cos - 1$$

when $n = 3$

$$x(3) = \frac{1}{4}\sum_{K=0}^{3} X(K)\, e^{j2\pi K 2/4}$$

$$= \frac{1}{4}\{X(0)\, e^{j2\pi 02/4} + X(1) e^{j2\pi 12/4} + X(2) e^{j2\pi 22/4} + X(3) e^{j2\pi 32/4}$$

$$= \frac{1}{4}(0 + 4e^{j3\pi/2} + 0 + 0)$$

$= e^{j3\pi/2}$

$= -j$

Exam-4 $x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)\, e^{j2\pi kn/N}$

$= \frac{1}{N} W_N^{*kn} X_N$

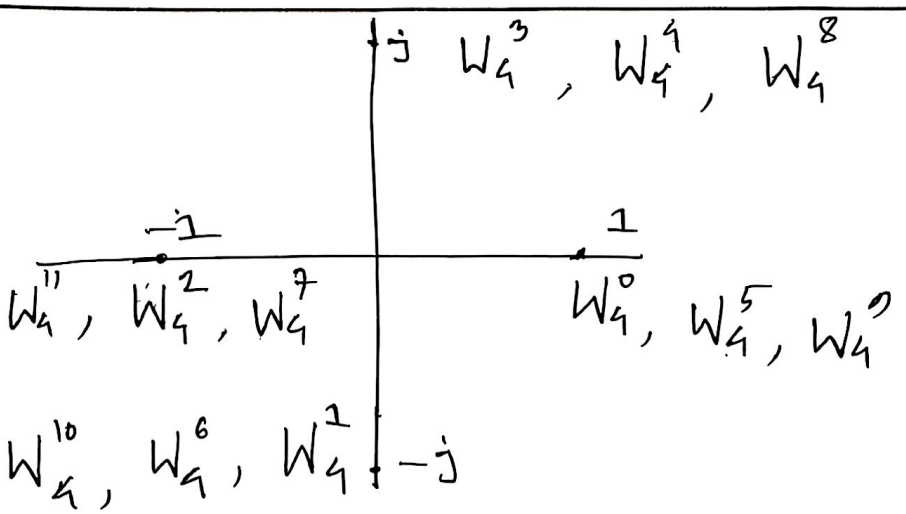
$= \frac{1}{4} W_4^{*kn} X_4$

$= \frac{1}{4}\begin{bmatrix} 1 & j & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}\begin{bmatrix} 0 \\ 4 \\ 0 \\ 0 \end{bmatrix}$

$= \frac{1}{4}\begin{bmatrix} 4 \\ 4j \\ -4 \\ -4j \end{bmatrix}$

$= \begin{bmatrix} 1 \\ j \\ -1 \\ -j \end{bmatrix}$

Twiddle factor



$$\begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W_4^6 \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix}$$

Lecture - 5

29.10.2026

(2.) Perform DFT of the following sequence :

$$x(n) = \{1, -1, 1, -1, 1, -1, 1, -1\}$$

Sol^n:

We know,

$$X(K) = \sum_{n=0}^{N-1} x(n)\, e^{-j2\pi kn/N}$$

$$\Rightarrow X(K) = \sum_{n=0}^{7} x(n)\, e^{-jkn2\pi/8}$$

$$\therefore X(K) = \sum_{n=0}^{7} x(n)\, e^{-jkn\pi/4}$$

When, $K=0$,

$$X(0) = \sum_{n=0}^{7} x(n) e^{0} = \sum_{n=0}^{7} x(n)$$

$$= 1 - 1 + 1 - 1 + 1 - 1 + 1 - 1$$

$$= 0$$

When $K=1$,

$$X(K) = \sum_{n=0}^{7} x(n)\, e^{-jn\pi/4}$$

$$= x(0) e^{-j0\pi/4} + x(1) e^{-j\pi/4} + x(2) e^{-j2\pi/4}$$
$$+ x(3) e^{-j3\pi/4} + x(4) e^{-j4\pi/4} + x(5) e^{-j5\pi/4}$$
$$+ x(6) e^{-j6\pi/4} + x(7) e^{-j7\pi/4}$$

$$= x - \frac{1}{\sqrt{2}} + j\frac{1}{\sqrt{2}} - j + \frac{1}{\sqrt{2}} + j\frac{1}{\sqrt{2}} - x + \frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}} + j - \frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}$$

$$= 0$$

when,

$K=2$, $X(2) = \sum_{n=0}^{7} x(n)\, e^{-j\cdot 2\pi n/4\cdot 2}$

$$= x(0)\, e^{-j2\pi 0/2} + x(1) e^{-j2\pi/2} + x(2) e^{-j2\pi/2} + x(3) e^{-j3\pi/2} + x(4) e^{-j4\pi/2} + x(5) e^{-j5\pi/2} + x(6) e^{-j6\pi/2} + x(7) e^{-j7\pi/2}$$

$$= x + j - x - j + x + j - x - j$$

$$= 0$$

when $K=3$,

$$X(K) = \sum_{n=0}^{7} x(n)\, e^{-j n 3\pi/4}$$

$$= x(0) e^{-j0\cdot 3\pi/4} + x(1) e^{-j3\pi/4} + x(2) e^{-j2\cdot 3\pi/4} + x(3) e^{-j3\cdot 3\pi/4} + x(4) e^{-j4\cdot 3\pi/4} + x(5) e^{-j5\pi\cdot 3/4} + x(6) e^{-j6\cdot 3\pi/4} + x(7) e^{-j7\pi\cdot 3/4}$$

$= 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}j + j - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}j - 1 - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}j$

$-j + \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}j$

$= 0$

when $K=4$,

$$X(4) = \sum_{n=0}^{7} x(n) \cdot e^{-j n \pi 4/4}$$

$$= \sum_{n=0}^{7} x(n) \cdot e^{-j n \pi}$$

$= x(0)e^{-j0\pi} + x(1)e^{-j\pi} + x(2)e^{-j2\pi} + x(3)e^{-j3\pi}$

$+ x(4)e^{-j4\pi} + x(5)e^{-j5\pi} + x(6)e^{-j6\pi} + x(7)e^{-j7\pi}$

$= 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1$

$= 8$

when $K=5$,

$$X(5) = \sum_{n=0}^{7} x(n) e^{-j n \pi 5/4}$$

$= x(0)e^{-j0\pi 5/4} + x(1)e^{-j\pi 5/4} + x(2)e^{-j2\pi 5/4}$

$+ x(3)e^{-j3\pi 5/4} + x(4)e^{-j4\pi 5/4}$

$+ x(5)e^{-j5\pi 5/4} + x(6)e^{-j6\pi 5/4} + x(7)e^{-7\pi 5/4}$

$= 1 + \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}j - j - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}j - 1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}j$

$+ j + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}j$

$= 0$

when $K=6$,

$$X(6) = \sum_{n=0}^{7} x(n)\, e^{-j\pi n 6/4} = \sum_{n=0}^{7} x(n)\, e^{-j n 3\pi/2}$$

$$= x(0)e^{-j0\,3\pi/2} + x(1)e^{-j3\pi/2} + x(2)e^{-j23\pi/2} + x(3)e^{-j33\pi/2} + x(4)e^{-j43\pi/2} + x(5)e^{-j53\pi} + x(6)e^{-j63\pi/2} + x(7)e^{-j73\pi/2}$$

$$= -j - x + j + x - j - x + j$$

$$= 0$$

when $K=7$,

$$X(7) = \sum_{n=0}^{7} x(n)\, e^{-j n 7\pi/4}$$

$$= x(0)e^{-j07\pi/4} + x(1)e^{-j7\pi/4} + x(2)e^{-j27\pi/4} + x(3)e^{-j37\pi/4} + x(4)e^{-j47\pi/4} + x(5)e^{-j57\pi/4} + x(6)e^{-j67\pi/4} + x(7)e^{-j77\pi/4}$$

$$= x - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}j + y + \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}j - x + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}j - y$$

$$\div \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}j$$

$$= 0$$

$$\therefore X(K) = \{0, 0, 0, 0, 8, 0, 0, 0\}$$

Lecture-6

01.11.2026

Q. Perform IDFT of the following sequence : $X(K) = \{0, 0, 0, 0, 8, 0, 0, 0\}$

Soln:

We know,

$$x(n) = \frac{1}{N} \sum_{K=0}^{N-1} X(K) e^{j 2\pi nk/N}$$

$$= \frac{1}{8} \sum_{K=0}^{7} X(K) e^{j\pi nk/4}$$

When $n = 0$,

~~$x(0) = \frac{1}{8} \sum_{K=0}^{7} X(0) e^{j\pi \cdot 0 \cdot 0/4} +$~~

$$x(0) = \frac{1}{8} \{ X(0) e^{j\pi \cdot 0 \cdot 0/4} + X(1) e^{j\pi \cdot 0 \cdot 1/4} +$$
$$X(2) e^{j\pi 0.2/4} + X(3) e^{j\pi 0.3/4} +$$
$$X(4) e^{j\pi 0 4/4} + X(5) e^{j\pi 05/4} + X(6) e^{j\pi 06/4}$$
$$+ X(7) e^{j\pi 07/4} \}$$

$$= \frac{1}{8}\{0+0+0+0+8+0+0+0\}$$

$$= 1$$

When $n=1$,

$$x(01) = \frac{1}{8}\{X(0)e^{j\pi 01\pi/4} + X(1)e^{j\pi 11/4} + X(2)e^{j\pi 21/4}$$
$$+X(03)e^{j\pi 31/4} + X(4)e^{j\pi 41/4} +$$
$$X(5)e^{j\pi 51/4} + X(6)e^{j\pi 61/4} + X(7)e^{j\pi 71/}$$

$$= \frac{1}{8}(-8)$$

$$= -1$$

Lecture-7
09-11-2026

Q. Perform DFT of the following sequence:

$$x(n) = \{1, -1, 1, -1, 1, -1, 1, -1\}$$

Sol^n: We know,

$$X(K) = \sum_{n=0}^{N-1} x(n)\, e^{-j 2\pi n K/N}$$

$$= W_N^{kn} x_N = W_8^{kn} x_8$$

Now,

$$\begin{bmatrix} W_8^0 & W_8^0 & W_8^0 & W_8^0 & W_8^0 & W_8^0 & W_8^0 & W_8^0 \\ W_8^0 & W_8^1 & W_8^2 & W_8^3 & W_8^4 & W_8^5 & W_8^6 & W_8^7 \\ W_8^0 & W_8^2 & W_8^4 & W_8^6 & W_8^8 & W_8^{10} & W_8^{12} & W_8^{14} \\ W_8^0 & W_8^3 & W_8^6 & W_8^9 & W_8^{12} & W_8^{15} & W_8^{18} & W_8^{21} \\ W_8^0 & W_8^4 & W_8^8 & W_8^{12} & W_8^{16} & W_8^{20} & W_8^{24} & W_8^{28} \\ W_8^0 & W_8^5 & W_8^{10} & W_8^{15} & W_8^{20} & W_8^{25} & W_8^{30} & W_8^{35} \\ W_8^0 & W_8^6 & W_8^{12} & W_8^{18} & W_8^{24} & W_8^{30} & W_8^{36} & W_8^{42} \\ W_8^0 & W_8^7 & W_8^{14} & W_8^{21} & W_8^{28} & W_8^{35} & W_8^{42} & W_8^{49} \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 1 \\ -1 \\ 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \frac{1}{\sqrt{2}}-j\frac{1}{\sqrt{2}} & -j & -\frac{1}{\sqrt{2}}-j\frac{1}{\sqrt{2}} & -1 & -\frac{1}{\sqrt{2}}+j\frac{1}{\sqrt{2}} & j & \frac{1}{\sqrt{2}}+j\frac{1}{\sqrt{2}} \\ 1 & -j & -1 & j & 1 & -j & -1 & j \\ 1 & -\frac{1}{\sqrt{2}}-j\frac{1}{\sqrt{2}} & j & \frac{1}{\sqrt{2}}-j\frac{1}{\sqrt{2}} & -1 & \frac{1}{\sqrt{2}}+j\frac{1}{\sqrt{2}} & -j & -\frac{1}{\sqrt{2}}+j\frac{1}{\sqrt{2}} \\ 1 & & & & & & & \\ 1 & & & & & & & \\ 1 & & & & & & & \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \frac{1}{\sqrt{2}}-j\frac{1}{\sqrt{2}} & -j & -\frac{1}{\sqrt{2}}-j\frac{1}{\sqrt{2}} & -1 & -\frac{1}{\sqrt{2}}+j\frac{1}{\sqrt{2}} & j & \frac{1}{\sqrt{2}}+j\frac{1}{\sqrt{2}} \\ 1 & -j & -1 & j & 1 & -j & -1 & j \\ 1 & -\frac{1}{\sqrt{2}}-j\frac{1}{\sqrt{2}} & j & \frac{1}{\sqrt{2}}-j\frac{1}{\sqrt{2}} & -1 & \frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}j & -j & -\frac{1}{\sqrt{2}}+j\frac{1}{\sqrt{2}} \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -\frac{1}{\sqrt{2}}+j\frac{1}{\sqrt{2}} & j & \frac{1}{\sqrt{2}}+j\frac{1}{\sqrt{2}} & -1 & \frac{1}{\sqrt{2}}-j\frac{1}{\sqrt{2}} & -j & -\frac{1}{\sqrt{2}}-j\frac{1}{\sqrt{2}} \\ 1 & j & -1 & -j & 1 & j & -1 & -j \\ 1 & \frac{1}{\sqrt{2}}+j\frac{1}{\sqrt{2}} & -j & -\frac{1}{\sqrt{2}}+j\frac{1}{\sqrt{2}} & -1 & -\frac{1}{\sqrt{2}}-j\frac{1}{\sqrt{2}} & j & \frac{1}{\sqrt{2}}-j\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \\ 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1-1+1-1+1-1+1-1 \\ 1-\frac{1}{\sqrt{2}}+j\frac{1}{\sqrt{2}}-j+\frac{1}{\sqrt{2}}+j\frac{1}{\sqrt{2}}-1+\frac{1}{\sqrt{2}}-j\frac{1}{\sqrt{2}}+j-\frac{1}{\sqrt{2}}-j\frac{1}{\sqrt{2}} \\ 1+j-1-j+1+j-1-j \\ 1+\frac{1}{\sqrt{2}}+j\frac{1}{\sqrt{2}}+j-\frac{1}{\sqrt{2}}+j\frac{1}{\sqrt{2}}-1-\frac{1}{\sqrt{2}}-j\frac{1}{\sqrt{2}}-j+\frac{1}{\sqrt{2}}-j\frac{1}{\sqrt{2}} \\ 1+1+1+1+1+1+1+1 \\ \vdots \\ \vdots \\ \vdots \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 8 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$\therefore X(K) = \{0, 0, 0, 0, 8, 0, 0, 0\}$ (Ans)

Q. Perform IDFT of the following sequence:

$X(K) = \{0, 0, 0, 0, 8, 0, 0, 0\}$

Soln: We know,

$$x(n) = \frac{1}{8N} \sum_{K=0}^{N-1} X(K) e^{j 2\pi nK/N}$$

$$= \frac{1}{N} W_N^{*Kn} X_N$$

$$= \frac{1}{8} W_8^{*Kn} X_8$$